Mortality Patterns: Retrospective Estimation and scenario-based forecasts- **GitHub Documentation**

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Introduction & Overall Description

The GitHub repository 'mortality_estimation' has been created to examine mortality rates in different countries. The main objectives of this study are to:

- Project or **estimate the crude death** rate using measured values of various predictor variables, referred as the **retrospective part**,
- **forecast scenarios** for the next five months based on the predictions of these predictor variables, referred as the **forecasting part**.

To achieve these two main objectives, the GitHub repository is organised into **10** distinct folders, as shown in Figure 1. The following section will first guide you through the libraries and versions of R required for successful execution of the different steps. After that, we'll dive into a detailed exploration of the different folders, most of which are made up of four main parts: data, results, visualisation, and code.

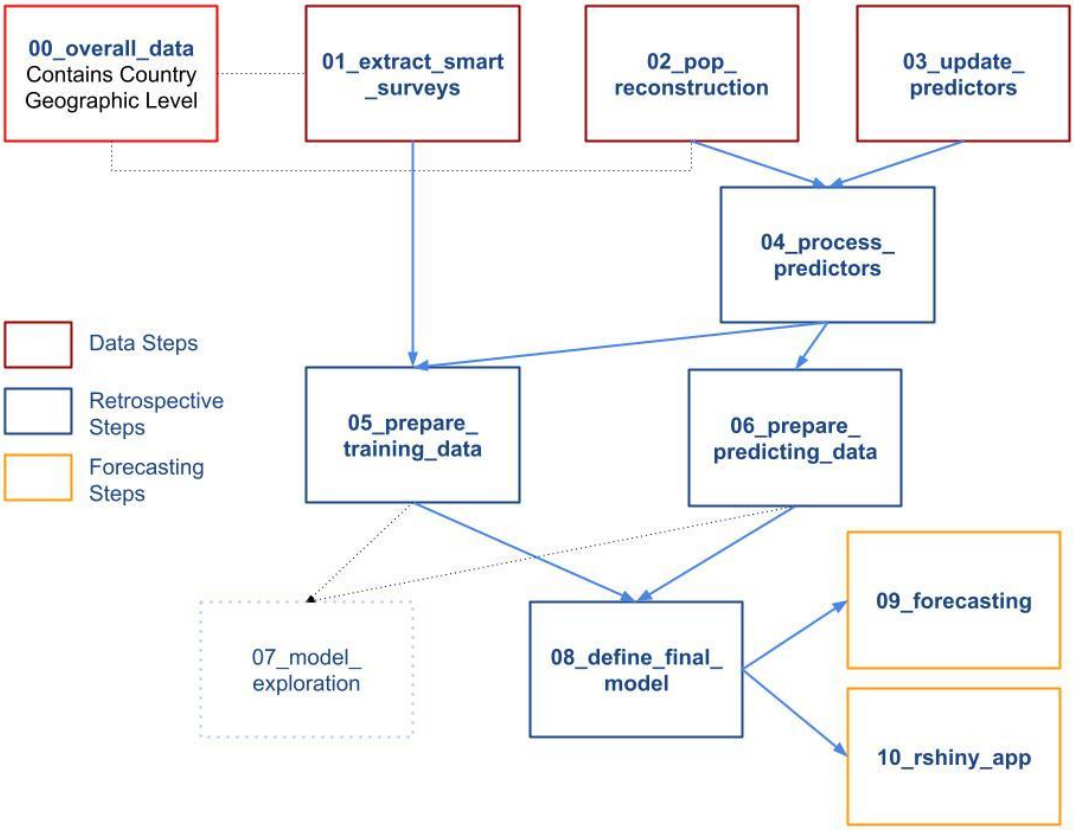

Figure 1. Visualization of the GitHub Structure

Note: If you only wish to view and experiment with the **RShiny App** for forecasting scenarios, you have two options. The first is to use the GitHub repository available in the following link: https://github.com/afyac/Forecasting_Scenario_App or to refer to the next section (Packages, libraries, and versions) to ensure that you have the necessary packages to launch the RShiny app. Then go to the folder entitled "10_rshiny_app" and follow the instructions provided in the following video: https://lshtm.cloud.panopto.eu/Panopto/Pages/Viewer.aspx?id=7ea184cf-2028-4554-867a-b0d901059162

Packages, Libraries & Versions

To run the RShiny code successfully, make sure you have access to the latest version of RStudio (R version 4.2.2). You can follow the instruction in this web page https://posit.co/download/rstudio-desktop/ if you do not have any R or RStudio version. Next, download this GitHub repository to your local device and double-click (or open) the 'MortalityEstimation.Rproj' file using RStudio.

After completing the above steps, download the required packages and load the necessary libraries. To do this, open and run the file entitled 'First_Run_Packages_Libraries.R'. Once you have run this file, you will be ready to launch all the code provided and the RShiny application.

It's important to **note** that we are utilizing the glmmTMB libraries, and the latest version may conflict with the most recent version of the Matrix package. Therefore, ensure that you have version 1.6-4 of the Matrix package. If you have any uncertainties, run the following R code line in the RStudio Console: `install.packages("Matrix", dependencies = TRUE, type='source')`

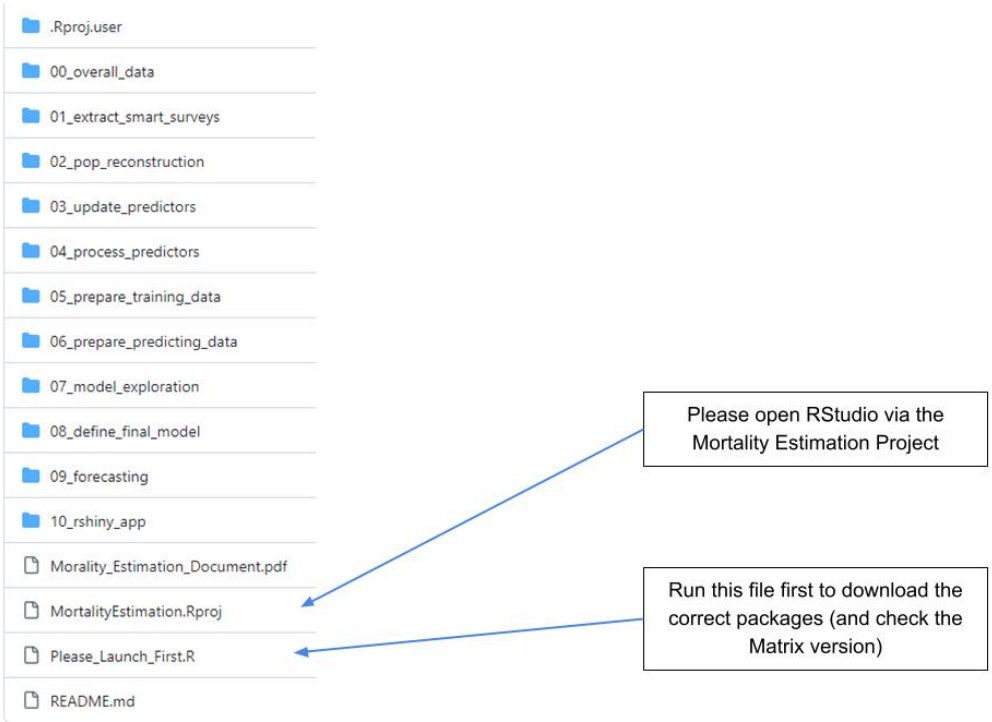

Figure 2. Steps to open the Project and Upload the right packages.

Mortality Estimation Steps

The GitHub repository consists of 10 separate folders. Each folder can be run independently, but its input or output may depend on other folders, as shown in Figure 1. To understand the main purpose and objective of this repository, please read the reports and annexes contained in the "11_reports" folder. These reports were produced for the Somalia mortality estimates. The following explanations are more specific to the code and do not explain in depth the different datasets or country specification.
Please note that each of the 10 dossiers has its own documentation which is more specific to the methodology used for each folder and which explains all the different functions step by step.

Step 1: Data Steps

The data processing stages consist of four separate folders, labelled 00 to 03, each tailored to the specific aspects of the country. The Somalia data is available in this GitHub repository.

1. **00_overall_data**: The initial folder focuses on datasets related to the geographical context of the country. Specifically, access to administrative levels 0, 1 and 2, as well as a list of different settlements or cities, is crucial for updating and processing SMART surveys and predictor variables.

2. **01_extract_smart_survey**: The second folder, 01_extract_smart_survey, contains four subfolders. The "data" subfolder contains the SMART surveys collected, while the "output" subfolder contains the cleaned metadata and data extracted from these surveys. The main purpose of this folder is to clean the SMART surveys, extracting information on mortality at the level of a small area (administrative level 2) over a specified period.

3. **02_pop_reconstruction**: The third file, 02_pop_reconstruction, is used when the population of a country is unknown in both time and space, and limited information is available (for example, the number of displaced people, refugees, or demographic data for a specific period). The output of this file provides population data classified by period and by administrative level 2.

4. **03_update_predictors**: The fourth folder, 03_update_predictors, contains different predictors essential for estimating the crude death rate. This folder and its associated code are specific to each country and require meticulous updating whenever new predictors are introduced.

At the end of the data collection process, three key results should be available: SMART survey data, demographic data and predictor variables classified by period and administrative level 2.

Step 2: Retrospective Steps

The retrospective process comprises five distinct files labelled 04 to 08, applicable to all countries, based on the results generated in the previous data stages.

1. **04_process_predictors**: In this initial folder, labelled **04_process_predictors**, the predictors undergo transformations, normalisation, classification, shifting and categorisation to ensure consistency in subsequent analytical steps. At the same time, population values are incorporated into the predictors during this phase.

2. **05_prepare_training_data**: The second folder, referred to as 05_prepare_training_data, involves the joining of administrative level 2 and temporal aspects between the SMART survey data (derived from folder 01_extract_smart_surveys) and the predictive variables (from folder 04_process_predictors). This process leads to the creation of training datasets, illustrated in Figure 3.

3. **06_prepare_testing_data**: In the third folder, 06_prepare_testing_data, a review is performed to check the availability of all periods and level 2 administrative data in the output of folder 04_process_predictors. In addition, visualisations are generated during this step.

4. **07_model_exploration**: The fourth folder, 07_model_exploration, serves as a testing ground for cross-validation, feature selection and models in order to identify the optimal model for estimating crude mortality rates. The exploration involves the use of the training datasets (output from folder 05_prepare_training_data) and the test dataset (output from folder 06_prepare_testing_data).

5. **08_define_final_model**: The last folder in this section, 08_define_final_model, uses the best model identified in folder 07_model_exploration. With this model, the final prediction of crude mortality rates is made, and counterfactuals are calculated. The final model is saved as output in this folder.

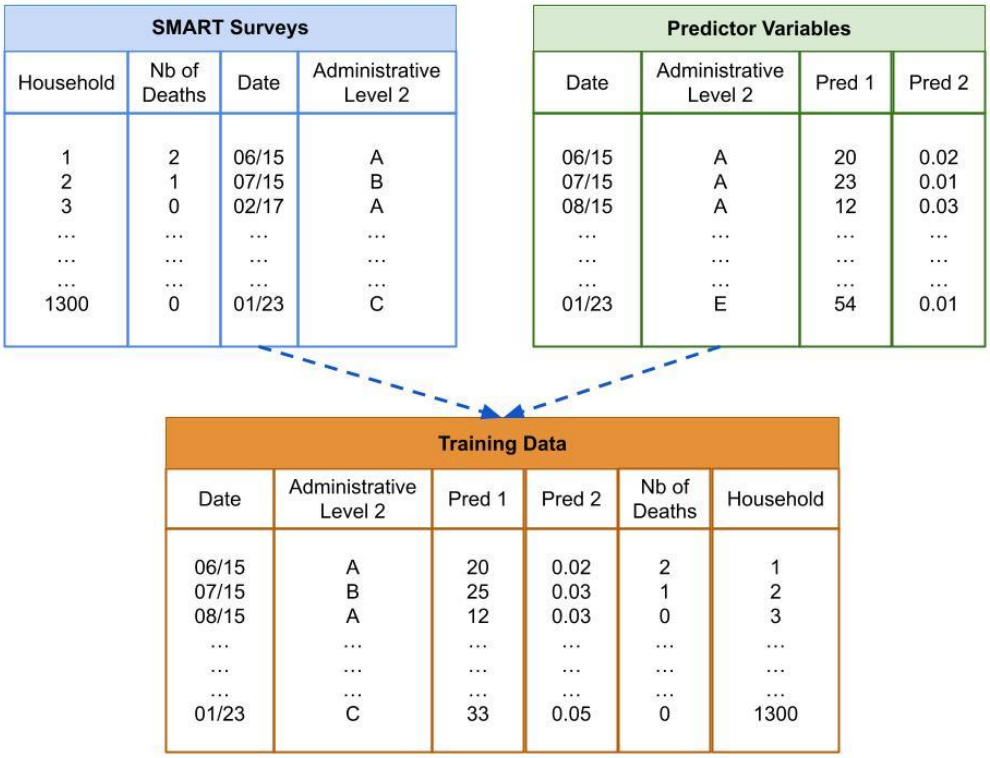

Figure 3. Construction of the training data.

Step 3: Forecasting Steps

The forecasting phase comprises two separate files. The first folder is dedicated to generating forecast figures for various reports, while the second focuses on creating an interactive app. Although these folders operate independently of each other, they share common input data. Specifically, to generate scenarios for forecasting crude mortality rates, the essential information needed are: the country's administrative levels (output from folder 00_overall_data), the predictor variables (output from folder 06_prepare_testing_data) and the final retrospective model (output from folder 08_define_final_model).

In the forecasting process, the initial step is to predict various predictor variables. Then, the crude mortality rate is estimated using these predicted predictors and the model defined in the 08_define_final_model file. Steps are explained in Figure 4. As a result, folder 09_forecasting produces the final values and visualisation graphs to be included in the report. In parallel, folder 10_rshiny_app contains an interactive application which allows the user to easily generate various forecast scenarios. A more in-depth exploration of the RShiny application is provided in the next section.

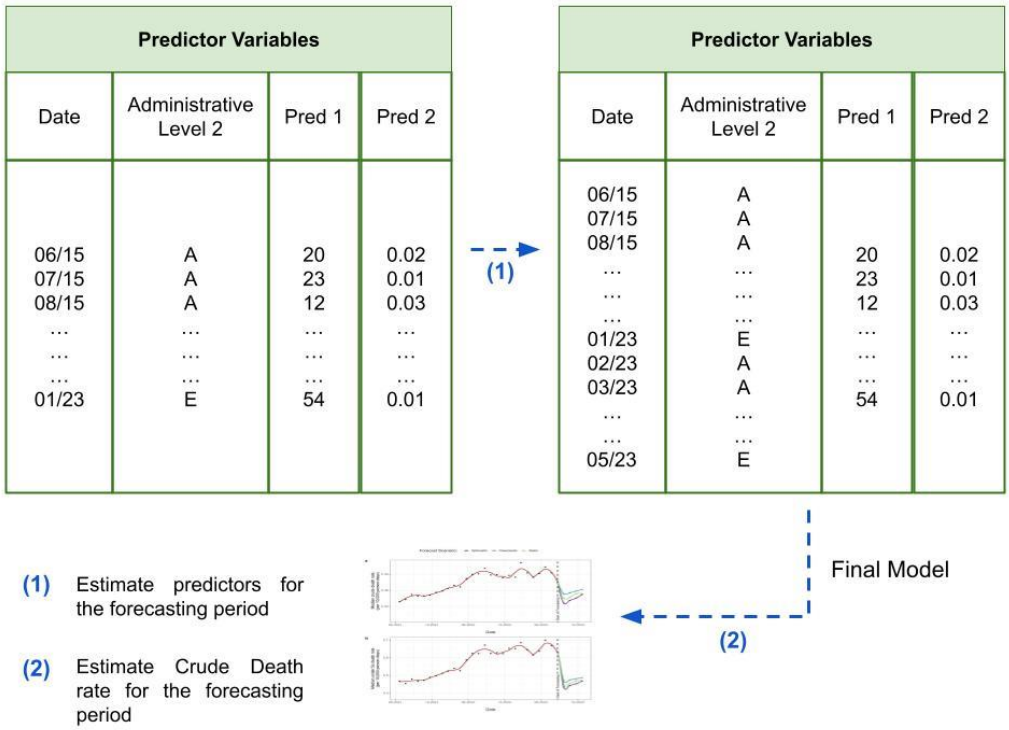

Figure 4. Forecasting Steps

RShiny App

The RShiny application contains the "forecasts" section, with a user-friendly interface to improve visualisation. This part can be used independently of the other folders. This is why you can find the required data either in the various folders (00_overall_data, 06_preparing_data or 08_define_final_model), or directly in the 10_rshiny_app folder in the data folder. To run the app, please read appendix 3 in the 11_report folder, the documentation in the 10_rshiny_app > documentation folder and/or follow the instructions in the following video: https://lshtm.cloud.panopto.eu/Panopto/Pages/Viewer.aspx?id=7ea184cf-2028-4554-867a-b0d901059162.